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Smooth Maps

1.1 Smooth Functions and Smooth Maps

1.1.1 Smooth Maps Between Manifolds

Definition 1.1.1 ► Smooth Maps Between Manifolds

Let M, N be smooth manifolds, and let $F : M \rightarrow N$ be any map. We say that F is a **smooth map** if for every $p \in M$, there exist smooth charts (U, φ) containing p and (V, ψ) containing $F(p)$ such that $F(U) \subseteq V$ and the composite map $\psi \circ F \circ \varphi^{-1}$ is smooth from $\varphi(U)$ to $\psi(V)$.

Remark.

- By taking $N = V = \mathbb{R}^k$ and $\psi = \text{Id} : \mathbb{R}^k \rightarrow \mathbb{R}^k$, we get the definition of smoothness of real-valued or vector-valued functions.